

Central tendency

mean
median

mean

$$x_1, \dots, x_n \quad \left| \quad \begin{array}{l} \text{mean} \\ \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \end{array} \right.$$

Value	freq
x_1	f_1
x_2	f_2
$\vdots$	$\vdots$
x_k	f_k
total	n

$$\bar{x} = \frac{1}{n} \sum_{i=1}^k x_i f_i$$

$$n = \sum_{i=1}^k f_i$$

Class	freq
9.05 - 16.55	1
16.55 - 24.05	5
total	6

Class-1

$$\text{class mark} = \frac{9.05 + 16.55}{2} = 12.8$$

Class-2

$$\text{class mark} = \frac{16.55 + 24.05}{2} = 20.3$$

$$\bar{x} = \frac{(12.8 \times 1) + (20.3 \times 5)}{6}$$

median

1, 2, 3, 4 $\bar{x} = 2.5$
1, 2, 3, 100 $\bar{x} = 26.5$

n: observations, arrange them in increasing order

median $\rightarrow$ middle most value.

n: odd median $\left(\frac{n+1}{2}\right)^{\text{th}}$ value

1, 2, 3, 4, 5

median = 3

n: even: any number between $\left(\frac{n}{2}\right)^{\text{th}}$ value and $\left(\frac{n}{2} + 1\right)^{\text{th}}$ value.

1, 2, 3, 4, 5, 6

by convention median $\frac{3+4}{2} = 3.5$

x value	freq	Cumulative freq
3	5	5
4	9	14
5	11	25
6	4	29
7	1	30
	30	

n=30

$$x_{15} = 5$$

$$x_{16} = 5$$

$$\text{median} = \frac{5+5}{2} = 5$$

Class interval	freq	Cumulative freq
149.55 - 149.55	1	1
149.55 - 154.55	3	4
154.55 - 159.55	24	28
total	28	86

$$P = \frac{1}{3}$$

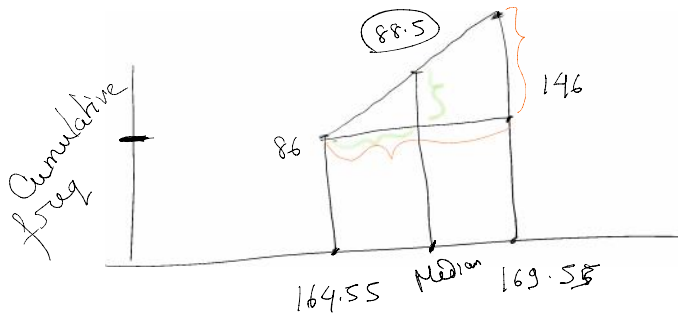
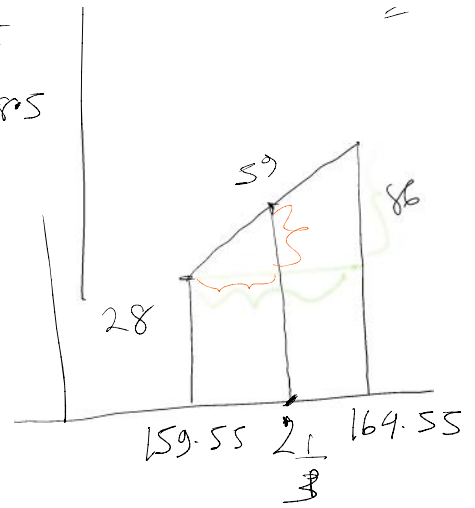
$$n \cdot P = 59.6$$

$$\frac{n}{2} = \frac{177}{2}$$

$$= 25.6$$

149.55 - 154.55	3	4
154.55 - 159.55	24	28
159.55 - 164.55	58	86
164.55 - 169.55	60	146
169.55 - 174.55	27	173
174.55 - 179.55	2	175
179.55 - 184.55	2	177
	177	

$$\frac{n}{2} = \frac{177}{2} = 88.5$$



$$\frac{169.55 - 164.55}{146 - 86} = \frac{\text{Median} - 164.55}{88.5 - 86}$$

$$\text{Median} = 164.76$$

Quantile (of order p)
 Z_p ($0 < p < 1$).

Z_p - is the value such that a proportion p of the total number of given observations are less than or equal to it and proportion $(1-p)$ of the total number of observations will be greater than or equal to it.

$x_1, x_2, \dots, x_n$

$$Q = (n+1)p$$

if Q is integer, $Z_p = x_Q$

$$1 \leq Q \leq n \quad p = \frac{Q}{n+1} \quad Q = 2.3 \quad p = 3.7$$

if Q is not integer $r = [Q]$ $\left| \begin{array}{ll} Q = 2.3 & Q = 3.7 \\ [Q] = 2 & [Q] = 3 \end{array} \right.$

$$Z_p = x_0 + (Q - r)(x_{r+1} - x_r)$$

(1, 2) (3) (4, 5)

$\left(\frac{1}{3}\right) \leq \left(2\frac{1}{3}\right) < \left(\frac{2}{3}\right)$

$\phi = \frac{1}{3}$